



Trimble® Earthworks AR Camera

For SITECH Only

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AR camera install (optional)

The optional AR camera (VC510) introduces an augmented view of the work area and your attachment. This section describes:

- Supported operation
- The contents of the camera kit
- How to adapt the camera power wires
- How to assemble and attach the camera mount
- How to install the Ethernet switch
- How to connect and route the cables
- How to enable the camera in the Web Interface

For instructions on calibration and operation, refer to the *Using the AR Camera Guide*.

CAUTION – Maximum torque values specified in this guide are important. Avoid over-tightening bolts.

1.1 Supported operation

- The camera is available on excavator systems.
- The camera is supported on TD520 displays in any screen layout mode. For the best experience, assign the camera to a larger pane.
- The camera is *not* supported on BYOD displays.
- The camera is *not* supported for guidance using an HS410 heading sensor.

1.2 Camera kit hardware

To install the camera, you require:

- Camera (VC510) (P/N 121101-00)
- Ethernet switch (SD510) (P/N 117546-00)
- Camera add-on kit (P/N 160015-525):
 - Installation instructions
 - M12 extension cable (5 m)
 - 4-pin extension cable (1 m)
 - 4-pin extension cable (4 m)
 - SD510 harness

- Four pot magnets
- Four M6 x 40mm button head bolts
- Four M6 nuts
- Five magnetic cable tie mounts
- 15 cable ties
- Camera mount kit (P/N 194265-00)
- 4-pin Deutsch connector
- 4-pin Deutsch wedgelock
- 4-pin Deutsch seal
- 4-pin Deutsch backshell
- Two terminal pins

You may require additional Deutsch extension cables, depending on the installation layout.

The plastic camera bracket included in the kit is not required and can be recycled.

For Cat NGH series 07 machines, you require an adaptor cable (P/N 150835-010).

For Cat mini/micro machines, use the standard cable.

1.3 Connecting the power wires

The camera's cable, as provided, has two connections:

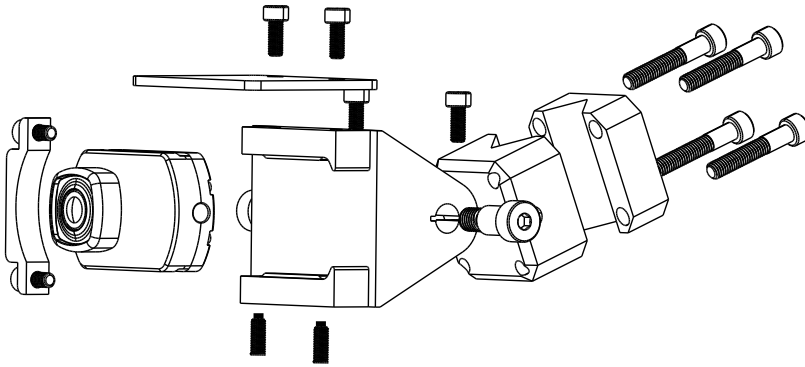
- An M12 connector
- A pair of bare power wires

Crimp pins to the bare power wires and insert them into the 4-pin Deutsch connector (provided in the SD510 harness kit):

- Red wire to pin 1 on the Deutsch connector
- Black wire to pin 4 on the Deutsch connector

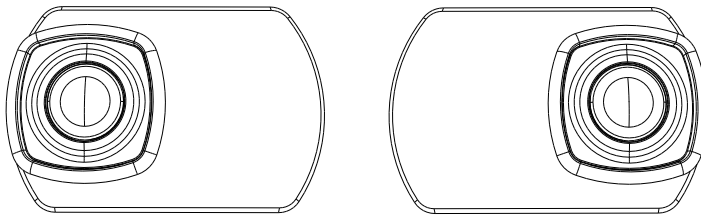
Pinout	Wire Color
1	1: Red
2	4: Black
3	
4	

1.4 Installing the camera

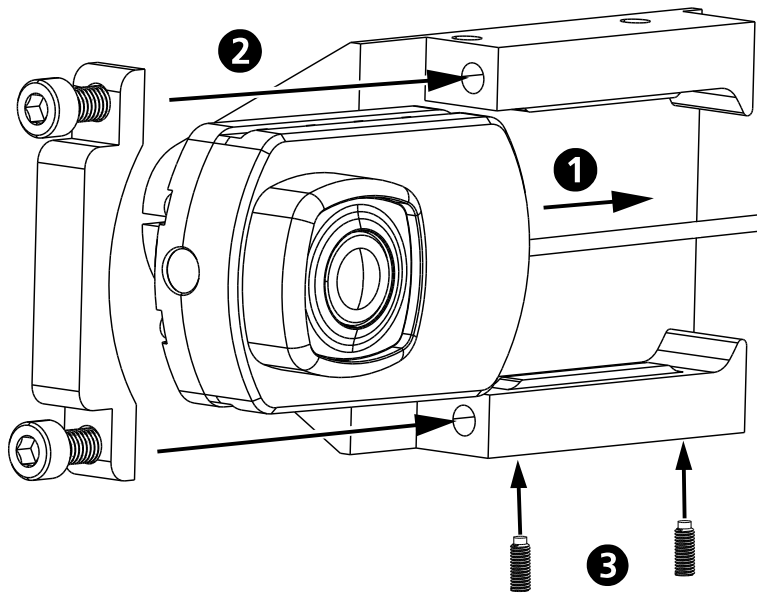


Mount the camera to a round bar on the cab:

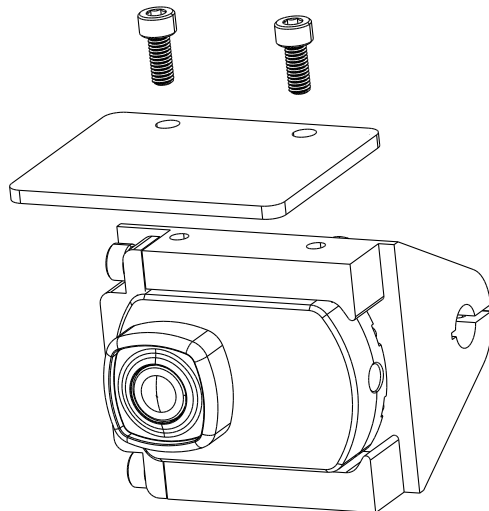
1. Select the camera mount location on the cab:
 - Select a position that provides an elevated, unobstructed view of the attachment. The recommended position is a vertical grab bar. Alternatively, you can mount to an existing bracket near the lights.
 - Select the camera orientation. The camera mounts horizontally.



2. Slide the camera cable-first into the slots on the bracket ❶. If necessary, invert the bracket to match your camera orientation.
3. Use two provided M5 bolts to attach the end plate which secures the camera ❷.
4. Apply Loctite to the two provided M5 x 0.8 mm set screws and install them to secure the camera to the side plate ❸.

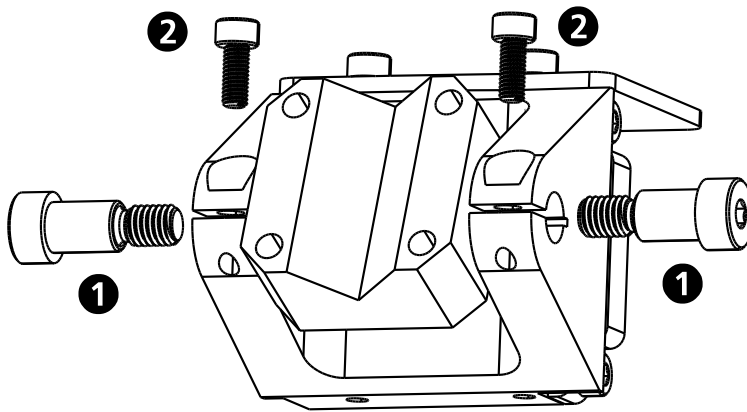


5. Align the weather shield plate with the holes in the top of the bracket and secure it with two provided M5 bolts.

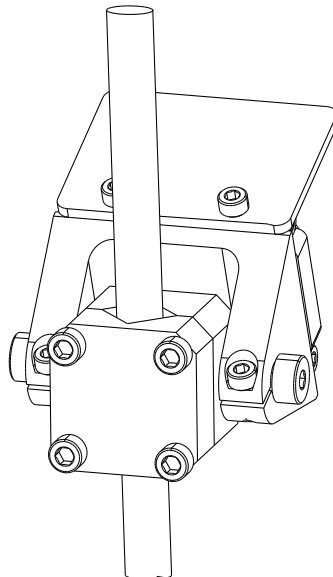


6. Place the larger V-block between the holes in the rear of the bracket and secure it with the two provided shoulder bolts ❶. Torque the bolts to 15 Nm (130 in-lbs).
7. Lower two provided M5 bolts into the vertical holes and secure ❷.

NOTE –Leave the M5 camera angle bolts finger tight. The camera angle will be adjusted and the bolts will be tightened in a later step.



8. Place the V-blocks around the mounting bar. Secure with four provided M6 x 35mm bolts.

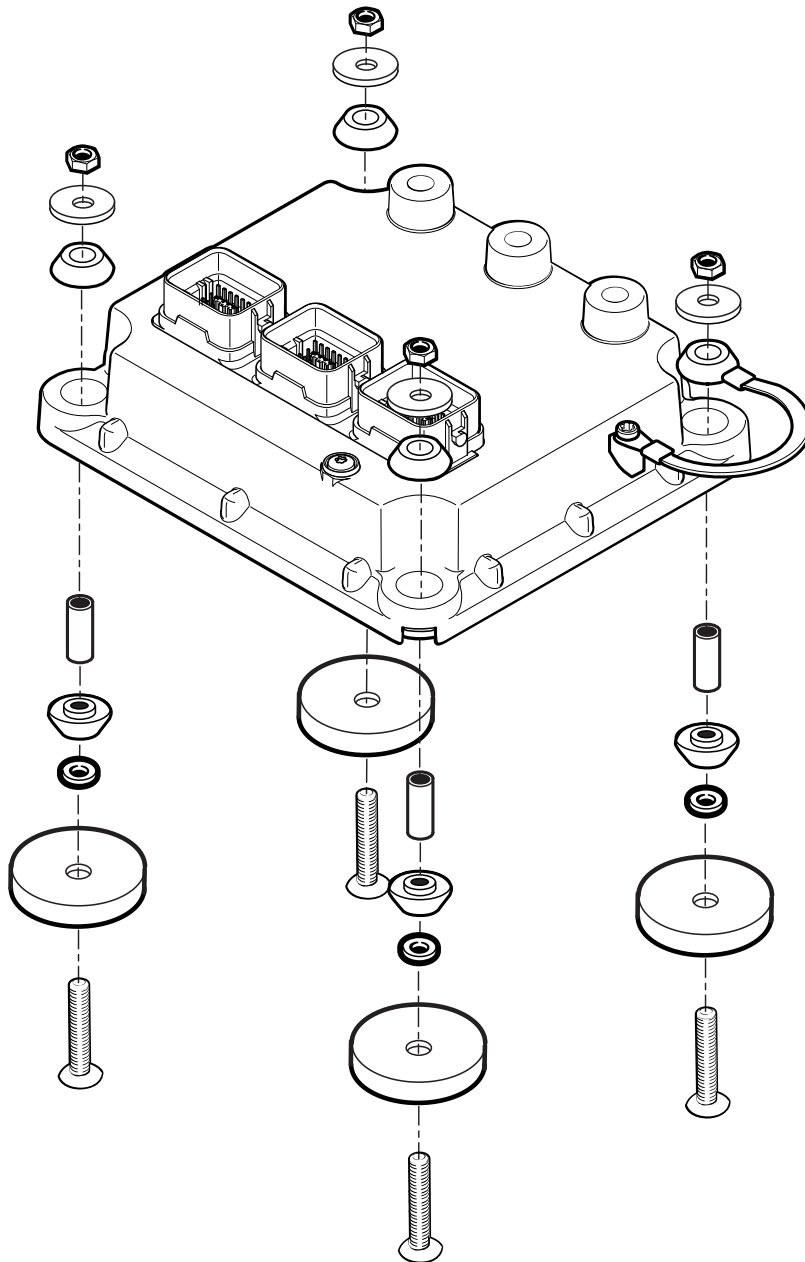


NOTE –Alternatively, you can exclude the smaller V-block and bolt straight into the larger V-block from an existing bracket near the lights on the machine.

1.5 Installing the Ethernet switch (SD510)

1. At each of the four corners of the Ethernet switch, install one rubber isolation mount from above and one from below.
2. Install a spacer inside each pair of isolation mounts.
3. Pass a bolt through each pot magnet.
4. Place a small washer over each bolt on top of the magnet.
5. Pass each bolt through the spacers and rubber isolation mounts on the Ethernet switch.
6. Place a large washer onto each bolt and then secure with a nut.
7. Use the magnets to attach the Ethernet switch to a metal surface.

Install the Ethernet switch within 2 m (7 ft) of the platform harness.



1.6 Connecting the cables

NOTE –The process for installing on a Cat NGH series 07 machine is described further down.

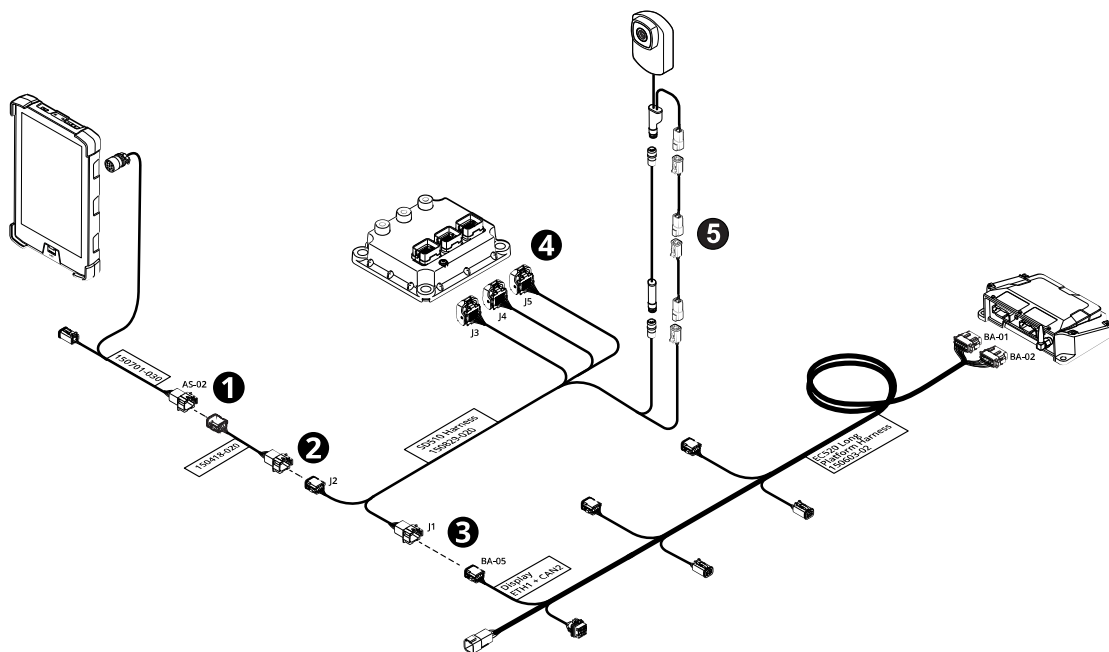
To install the cables, connect the SD510 harness to parts of the Trimble Earthworks system. The camera uses the 4-wire Ethernet. The 2-wire Ethernet remains unconnected.

1 AR camera install (optional)

1. Disconnect the extension cable from BA-05 on the platform harness (P/N 150603-02).
2. Connect BA-05 on the platform harness to the 8-pin male Deutsch connector on the SD510 harness **3**.
3. Connect the extension cable to the 8-pin female Deutsch connector on the SD510 harness (P/N 150829-020) **2**.

NOTE –Depending on the installation layout, you may prefer to install the SD510 harness at the other end of the extension cable **1**.

4. Attach the J3, J4, and J5 connectors on the SD510 harness to the 3 sockets on the Ethernet switch **4**.
5. Connect one end of the M12 extension cable to the 4-wire Ethernet connector on the SD510 harness.
6. Connect the other end of the M12 extension cable to the M12 connector on the camera cable.
7. Connect the 1 m and 4 m Deutsch extension cables between the camera's Deutsch connector (that you assembled, see [1.3 Connecting the power wires](#)) and the 4-pin Deutsch on the SD510 harness **5**.

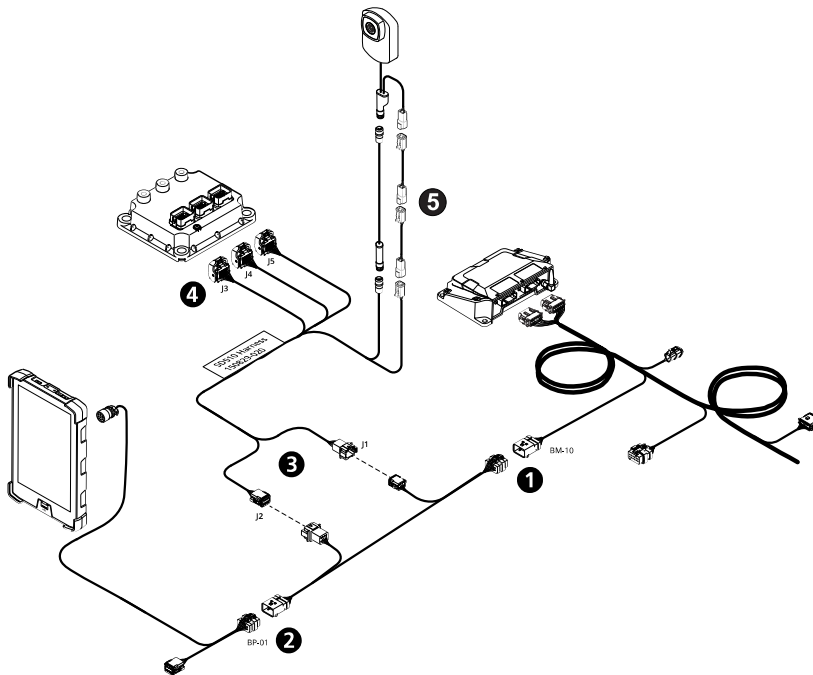


On Cat NGH series 07 machines, the installation requires an additional adaptor cable:

1. Locate connector BM-10 on the platform harness (P/N 150607-01). The connector is in the floor of the cab, behind the seat. To access it, remove the cup holder and plastic panel. The connector is not labeled.

1 AR camera install (optional)

2. Connect the female AMPSEAL connector on the adaptor cable (P/N 150835-010) to BM-10 **1**.
3. Connect the male AMPSEAL connector on the adaptor cable to BP-01 on the display cable **2**.
4. Connect the two 8-pin Deutsch connectors on the SD510 harness (P/N 150829-020) to the two Deutsch connectors on the adaptor cable **3**.
5. Attach the J3, J4, and J5 connectors on the SD510 harness to the 3 sockets on the Ethernet switch **4**.
6. Connect one end of the M12 extension cable to the 4-wire Ethernet connector on the SD510 harness.
7. Connect the other end of the M12 extension cable to the M12 connector on the camera cable.
8. Connect the 1 m and 4 m Deutsch extension cables between the camera's Deutsch connector (that you assembled, see [1.3 Connecting the power wires](#)) and the 4-pin Deutsch on the SD510 harness **5**.



1.7 Securing the cables

Where the extension cable runs along the machine body, secure it with the provided cable ties.

If the machine lacks clear fastening points, install the provided mount magnets along the outside of the machine body and then attach the cable to the magnets with cable ties.

1.8 Installing the license

To use the AR camera system, you require *Module License – Earthmoving Workflow Optimization*.

For information on installing licenses, refer to the *Configuration Guide* in the *Excavator Commissioning Manual*.

1.9 Checking the display's Ethernet setting

NOTE –If you have used the TD5x0 display with the simulator, this step is important.

If the TD5x0 display's Ethernet is set to Static IP, the camera feed will not display.

On a TD520 display, go to Settings > Ethernet > Configure Ethernet and ensure the Connection Type is set to DHCP.

On a TD540 display:

1. Swipe-up on the main screen to open the app menu.
2. Select Ethernet app.
3. Select Configure Ethernet.
4. Select DHCP.

1.10 Enabling the camera

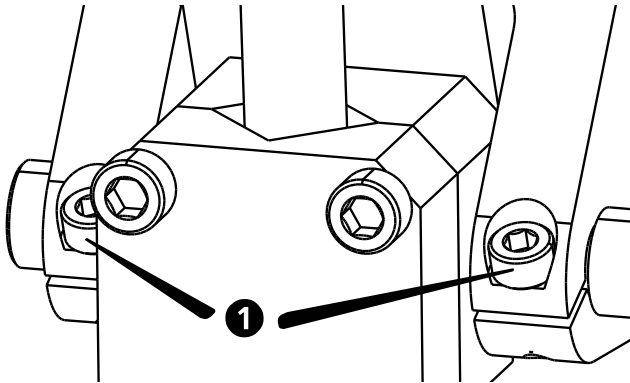
1. From the Web Interface, select Configure > Install Assistant.
2. Proceed through the Setup section.
3. On the Optional Devices screen, set the AR Camera toggle to on.
4. Select the camera orientation. The onscreen image shows the angle selected.
5. Complete the Setup section.
6. Select Monitor > Onboard Devices. Verify that the camera is connected correctly.

1.11 Calibrating the camera

Calibrate the camera. Refer to the *Trimble Earthworks Excavator Operator's Manual*.

NOTE –You can use the calibration screen to adjust the camera angle before setting the first set of points.

When the camera view is active on the operator interface, adjust the angle of the camera within the housing so it shows the work area. When the camera angle is correct, tighten the two camera angle bolts to 8 Nm (70 in-lbs) ①.



The bracket will remain adjustable by hand, but will not move due to operation.

1.12 Troubleshooting

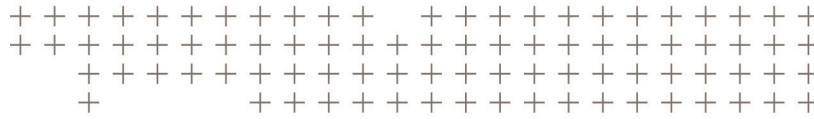
1.12.1 Disconnect harness when downgrading system software

If you downgrade the system to an earlier firmware version (for example, v1.11), you must disconnect the SD510 harness from the EC520 harness first. If the SD510 harness remains connected, machine files will fail to load.

1.12.2 Missing camera feed caused by Ethernet connection type

If the camera's feed does not show on the display, check the display's Ethernet connection setting.

On the display, go to Settings > Ethernet > Configure Ethernet and ensure the Connection Type is set to DHCP.



Notices

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